

Hemalatha N

hemanagarajan1603@gmail.com | +91 7305144267 | [linkedin.com/in/hemalatha-n-80964617a](https://www.linkedin.com/in/hemalatha-n-80964617a)

ABOUT ME

- Dedicated to unraveling complexities and empowering others.
- As a skilled debugger with a passion for understanding code, I thrive on solving technical challenges.
- My commitment to continuous learning drives me to excel in my projects and career.
- With a knack for breaking down problems and providing effective solutions, I'm an asset to any team.

Professional summary

- Ability to architect, implement, and test complex embedded systems telematics framework across various platforms.
- With a strong foundation in firmware, hardware, and software design, I am adept at driving projects from concept to commercialization.
- Experience in test-driven development, manufacturing processes, technical documentation, and delivering high-quality, high-volume products ensures successful project execution.

TECHNICAL SKILLS

Programming Languages: C/C++, Python, SQL.

Domain specific skills: C++ (11/14/17), Linux, QT, STL containers, Multi-threading, GRPC OOPS, GTest, RTOS, Protocol Buffers, Architecture (x86, arm),
Design Pattern, Modbus, Telematics, Exception Handling, ZeroMQ, Protocols (TCP/IP, UDP, CAN/J1939, SFTP, UART), System Design, Computer Vision, OpenCV,

Additional skills : Javascript, HTML, CSS, REACTjs, shell scripting.

Developer tools : Tools (Wireshark, Canalyzer, VSCode), Memory analyzers (GDB, Valgrind), Static analyzers (cppcheck, flaw finder), Git, Jenkins, Docker, Putty, Winscp, QTDesigner.

COURSEWORK: Data Structures, DBMS, Generative AI for everyone.

EDUCATION

SASTRA UNIVERSITY

Thanjavur, Tamil Nadu
MCA | CGPA: 7.4 | Grad July 2022

SSKV COLLEGE OF ARTS AND SCIENCE FOR WOMEN

Kanchipuram, Tamil Nadu
BSc Computer Science | CGPA: 8.9 | Grad July 2020

EXPERIENCE

TATA ELXSI | SENIOR ENGINEER

OCT 2022 – Nov 2025 | Trivandrum, Kerala

Project: Telematics

- Application layer development for telematics domain for off-road vehicle using C++ on Linux platform (LXC container).
- This project contains various environment with different architecture to provide solutions for the customer needs.

Roles and Responsibilities:

- Developing and maintaining software for modules like **MineStar**, **HPGPS**, **reports**, **schedulers**, **webpage** communication using thrift.
- Understanding **customer requirements** and providing solutions.
- Encoding, Decoding of Google Protobuf messages which act as **storage** medium.
- Verifying the code implemented by **unit testing** using Gtest, **static analysis** tools like flaw finder, cppcheck, creating test plan and performing **functional testing**.
- Creating applications on Linux platform and integrated to the workspace so that some Realtime data can be simulated which will be helpful for debugging (**Memory analyzer**)
- Creating **automation scripts using python** to test few features which reduce some manual efforts.

OPTIMIZED ELECTROTECH | SENIOR SOFTWARE ENGINEER

DEC 2025 – Present | Bangalore, Karnataka

Project: Surveillance System

- Development of software for advanced surveillance camera systems, including products like **WolfSight**, **NoctVision**, and **thermal imaging cameras**, deployed in defense environments.

Roles and Responsibilities:

- Design and implement software modules for camera control, image processing, and real-time video streaming using **C++**, **C#**, and **QT/QML**.
- Develop interfaces for camera systems and integrate with backend services using secure protocols. Implement GUI components for monitoring and control.
- **Performance Optimization:** Optimize image processing pipelines for **low latency** and **high-resolution video** in challenging environments (night vision, thermal imaging).